THABANG WASHINGTON MANYAMA

Alberton, Gauteng, South Africa · 069 833 0557 · thabangmanyama19@gmail.com
Portfolio: don4chnge.github.io · linkedin.com/in/thabang-manyama-593777228 · github.com/Don4chnge

PROFILE

Thabang Manyama, 21. Final-year BSc IT (Data Science) student who turns raw South African data into tested, deployed systems. Team lead of Ubufakazi, a live scam-forensics platform that won 2nd place at HackSecure 2026, and author of Valorian Intelligence, an ML monitoring library that caught a model failure input-only monitoring missed. Works end to end, from sourcing and validating institutional data to shipping a tool people can use. 14 module distinctions, 2× Academic Top Achiever at EDUVOS, three Microsoft certifications and podium finishes at two 2026 security hackathons.

TECHNICAL SKILLS

Languages Python, R, SQL, JavaScript

Machine learning scikit-learn, XGBoost, KMeans, logistic regression, random forests, feature engineering, time-ordered validation, drift detection and model monitoring

Statistics Hypothesis and permutation testing, PSI, Kolmogorov-Smirnov, chi-square, OLS, distributed-lag and fixed-effects regression, confidence intervals

Data pandas, NumPy, SciPy, statsmodels, SQL, PostgreSQL, SQLite, Microsoft Fabric, reproducible pipelines from raw institutional data

Visualisation D3.js, Streamlit, Power BI, Tableau, Matplotlib, Excel

Engineering Git and GitHub, GitHub Actions CI, pytest, Playwright, FastAPI, Render, Netlify, object-oriented design

Security & cloud SHA-256 evidence hashing, HMAC-signed APIs, two-factor authentication, Content Security Policy, Azure Machine Learning, Kubernetes fundamentals

SELECTED PROJECTS

Ubufakazi - Scam Evidence and Syndicate-Linking Platform
Oct 2026

2nd Place, HackSecure 2026 (CyberM8), Track 4 | Team lead · ubufakazi.onrender.com · github.com/Don4chnge/ubufakazi

- **Led a three-person team to 2nd place** by shipping a live platform that turns a scam victim's WhatsApp chats and payment slips into a tamper-evident, police-ready case file in under 60 seconds, in five South African languages.
- **Built the Telegram reporting bot and FastAPI backend** (Render, Neon PostgreSQL) with server-side HMAC keys and RFC 3161 trusted timestamps, so every evidence file can later be proven unchanged.
- **Designed a privacy-preserving "Scam Fingerprint"** from keyed HMAC-SHA256 identifiers and message-script similarity, which keeps a syndicate linked across stations and provinces even after it changes phone numbers and bank accounts.
- **Secured and tested it end to end:** two-factor sign-in, station isolation and a hash-chained audit log for police users, and a 70-case, five-language test pack in which every case returned the correct account, amount and scam type.

Valorian Intelligence - ML Observability Toolkit
2026

Open-source Python library and dashboard · valorian-intelligence.streamlit.app · github.com/Don4chnge/valorian-intelligence

- **Caught an 11.3% drop in model performance that input-only monitoring missed** by building a library that tracks both data drift and concept drift in deployed ML models.
- **Combined PSI, Kolmogorov-Smirnov and chi-square tests with a domain classifier** that detects multivariate shifts single-feature tests cannot see, and designed DriftScore, a 0-100 model health metric with documented weights.
- **Validated on six quarters of Stats SA labour-force microdata** (31,000 to 41,000 people per quarter): zero false alarms across five quarters, plus a hidden shift confirmed by permutation test ($p = 0.048$).
- **Shipped production-ready:** 78 automated tests, CI across Python 3.10 to 3.12, and a deployed Streamlit dashboard.

Hung Council Watch - 2026 Local Election Forecasting
2026

DIRISA Student Datathon Challenge (CSIR) | ML modelling and deployment, team of six · eduvos-hung-councils.netlify.app · GitHub repository

- **Flagged 62 of 72 hung councils on an election the model never saw (ROC-AUC 0.88)** and forecast about 77 hung councils across South Africa's 213 municipalities for 2026, using IEC results from 2000 to 2021.
- **Chose the model on evidence, not leaderboard score:** compared a baseline, two logistic regressions and tuned XGBoost with time-ordered validation, and kept the simpler model after XGBoost won ordinary cross-validation but lost on future elections.
- **Caught two errors in the official source data** (a 2016 turnout column that counted only the leading party's votes, and 2021 data cut off by Excel's row limit), and tested and rejected the team's first hypothesis with U-tests and fixed-effects models.

- **Deployed a public D3.js dashboard** with a risk map, swing scenarios and an election-day simulator that runs the trained model in the browser, guarded by 25 automated validation tests in CI.

Load Shedding's Economic Ripple Effect Jul 2026

Statistical impact study | Python, statsmodels, Power BI · github.com/Don4chnge/loadshedding-economy

- **Measured how Eskom load shedding moved South African retail sales** across 49 months (Apr 2022 to Apr 2026), sourcing raw data directly from the Eskom Data Portal and Stats SA (P6242.1).
- **Found a strong negative relationship** ($r = -0.649$, $p < 0.0001$, 95% CI -0.786 to -0.450) that held at lags of 0 to 4 months and in an OLS model controlling for the time trend.
- **Translated the statistics into business terms:** moving from a month with no load shedding to the worst month (3,213 MW) is linked to a 6.0 percentage-point swing in retail sales growth (95% CI 3.0 to 9.1).
- **Delivered a Power BI dashboard, a reproducible Jupyter pipeline and a research report** that states confounders and limitations up front.

Secure Session Guardian 2026

3rd Place, ITWeb Security Summit Hackathon | Account-takeover defence

- **Placed 3rd at the ITWeb Security Summit Hackathon** with a platform that detects session hijacking, token theft and account takeover through continuous session monitoring and AI-driven risk scoring, with adaptive controls that step up protection as risk rises.

HONOURS & AWARDS

- **2nd Place, HackSecure 2026 (CyberM8), Oct 2026.** Track 4: Digital Forensics Lite. Team lead, Ubufakazi.
- **3rd Place, ITWeb Security Summit Hackathon, 2026.** Secure Session Guardian.
- **2× Academic Top Achiever, EDUVOS, 2025 and 2026.** Top-performing student, BSc IT: Data Science, in two consecutive years.
- **Golden Key International Honour Society, 2025.** Invited membership for the top 15% of the academic year.

EDUCATION

BSc Information Technology: Data Science Feb 2024 - Present

EDUVOS | NQF Level 7, SAQA ID 120690 | 347 credits completed

- **14 distinctions across three years,** including 97% in Linear Algebra with Applications.
- **Year 1 (2024), 1 distinction:** Procedural Programming (89%).
- **Year 2 (2025), 6 distinctions:** IT Project Management (89%), Web Server Management (88%), Machine Learning Algorithms (85%), Mathematics 1B (81%), Artificial Intelligence Techniques (77%), Probability and Statistics (77%).
- **Year 3 (2026, in progress), 7 distinctions:** Linear Algebra with Applications (97%), Operating Systems (88%), 4IR Technologies (84%), Sensors Data Management, IoT Mining and Analytics (82%), Object Oriented Systems Analysis and Design (80%), R Programming for Data Science (80%), Research Design and Methodology (76%).
- **Further coursework:** Database Systems, Scientific Computing in Python, Data Structures and Algorithms in Python, Network Security, Cloud-Based Technologies, AI Ethics and Privacy, Project: Data Science.

EXPERIENCE & LEADERSHIP

Delegate, Youth 20 (Y20), G20 Mar 2025 - Nov 2025

- **Represented South African youth** in the G20 engagement process, advancing positions on youth empowerment, innovation and financial inclusion alongside delegates from member economies.

Participant, All4Youth Sub-Saharan Africa Leadership Programme Current

Leadership development in teamwork, communication, problem-solving and community impact

Retail Assistant, Sportscene, Sandton City Oct 2024 - Apr 2025

- **Served 50+ customers a day** in a high-traffic store while studying full time, and handled stock control and visual merchandising.

CERTIFICATIONS

- **Microsoft Certified: Fabric Data Engineer Associate (DP-700),** Aug 2026
- **Microsoft Certified: Power BI Data Analyst Associate (PL-300),** Apr 2026
- **Microsoft Certified: Azure Data Scientist Associate (DP-100),** Apr 2026
- **Kubernetes and Cloud Native Essentials** (course)